Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune 37

(An Autonomous Institute affiliated to Savitribai Phule Pune University,Accredited with 'A++' Grade by NAAC)

Department of Engineering, Sciences and Humanities, A. Y. 2024-25

Name of the Course: Calculus and Statistics (HS

Tutorial 12 on Unit 6-Random Variables:Bivariate distribution

. **Examples**

1. Consider the following bivariate distribution.

X \ Y	0	1	$P(X)$	For (X)
-1	0	$\frac{1}{3}$	$\frac{1}{3}$	
0	$\frac{1}{3}$	0	$\frac{1}{3}$	
1	0	$\frac{1}{3}$	$\frac{1}{3}$	
$P(Y)$	$\frac{1}{3}$	$\frac{2}{3}$	1	$\rightarrow$ For (Y)

1. Write the marginal distribution for X and Y.
2. Find $Cov(X,Y)$.
3. Find $\rho(X,Y)$
4. Write the conclusion about value of $\rho(X,Y)$.
5. **5.** Are the random variables Independent?

2. Let X and Y be independent random variables with the following distributions :

Find the joint distribution of X and Y.

X	1	2
$p(X=x_i)$	0.6	0.4

Y	5	10	15
$p(Y=y_j)$	0.2	0.5	0.3

3. A fair coin is tossed three times. Let X equals 0 or 1 according as a head or a tail occurs on the first toss. Let Y equals the total number of heads that occur in each toss.

a) Write the marginal distributions of X and Y.

b) Write the joint distribution of X and Y.

c) Find $Cov(X,Y)$

d) Find $\rho(X,Y)$. Comment on value of $\rho(X,Y)$.

e) Determine whether X and Y are independent.

4. Let X and Y be random variables with the joint distribution.

(a) Find the distribution of X and Y.

(b) Find Cov (X, Y).

Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune 37

(An Autonomous Institute affiliated to Savitribai Phule Pune University,Accredited with 'A++' Grade by NAAC)

Department of Engineering, Sciences and Humanities, A. Y. 2024-25

Name of the Course: Calculus and Statistics (HS

Tutorial 12 on Unit 6-Random Variables:Bivariate distribution

(c) Find $\rho(X, Y)$.

(d) Are X and Y independent random variables?

Y/X	-3	2	4	**Sum**
1	0.1	0.2	0.2	0.5
3	0.3	0.1	0.1	0.5
Sum	0.4	0.3	0.3	

5. A fair coin is tossed three times. Let X equal according as a head or a tail occurs on the first toss and let Y equal the total number of heads that occur.

(a) Find the distribution of X and Y.

(b) Find the joint distribution h of X and Y.

(c) Determine whether X and Y are independent.

(d) Find Cov (X, Y).

6. Let X be the random variable with the following distribution and let $Y = X^2$.

x	-2	-1	1	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

(a) Find the distribution of Y.

(b) Find the joint distribution of X and Y.

(c) Find Cov (X, Y) and $\rho(X, Y)$

(d) Determine whether X and Y are independent.

7. A fair coin is tossed four times. Let X denote the number of heads occurring and let Y denote the longest string of heads occurring.

(a) Determine the joint distribution of X and Y.

(b) Find Cov (X, Y) and $\rho(X, Y)$.